Indian Institute of Information Technology Kottayam
Department of Electronics and Communication Engineering
Organizes
Hands-on Electromagnetic Simulation Training on

Design and Analysis of Antennas and Microwave Devices and Circuits using HFSS
(Online Mode)

(20 October, 2023)

About IIIT Kottayam

Indian Institute of Information Technology Kottayam at Pala, Kerala is one among the IIITs that have been established as "Institution of National Importance" by MoE, Govt. of India under the ambit of IIIT PPP Act 2017. IIIT Kottayam campus is situated at Valavoor, Kottayam - a picturous location, spread over 53 acres of land - allotted by Govt. of Kerala in the southern part of the Country commonly referred to as God’s Own Country. B.Tech students are admitted against the JoSAA/CSAB counselling against the JEE main rank list. Institute offers 4 year B.Tech and B.Tech (Hons) programs in ECE, CSE and Cybersecurity. In addition, Institute also offers Ph.D programs in ECE, CSE and Mathematics as well as M.Tech programme in (1) AI & Data Science (2) Cyber Security for working professionals.

About Ansys/HFSS

Ansys HFSS is a 3D Electromagnetic (EM) Field Simulator for RF and Wireless Design. The software is used for designing and simulating high-frequency electronic products such as antennas, antenna arrays, RF or microwave components, high-speed interconnects, filters, connectors, IC packages and printed circuit boards. Engineers worldwide use Ansys HFSS software to design high-frequency, high-speed electronics found in communications systems, advanced driver assistance systems (ADAS), satellites, and internet-of-things (IoT) products. Improved antenna simulation is possible in larger structures, further reinforcing the HFSS ability to execute EM simulations over scale.

About Entuple Technologies

Entuple Technologies was founded by professionals with a combined experience of over 80 years in the Electronics Industry. The management team with its experience in different sectors such as Aerospace & Defence, Small & Medium Business, Research & Academia has joined together to build a world class team of Next Generation Solution Enablers in system design technologies. India being one of the emerging markets has been identified as the breeding ground of leading R&D initiatives in multiple domains. Entuple is committed to bridge the ever growing gap in the industry by bringing in expertise to meet technological challenges by introducing cutting edge platforms, tools and solutions. In the academic sector, Entuple is committed to bridge the growing gap between curriculum and advancements in the industry by providing effective tools and technologies.

About need for Simulation Training

Simulation modeling solves real-world problems efficiently. It provides analysis which is easily verified and communicated. The main reasons for performing electromagnetic (EM) simulation are to solve unintended EM interactions in the circuits/systems and ensure design meets performance specifications. Antenna simulation helps finding optimal antenna designs and reveals installed performance and interference in real-world environments. Engineers responsible for integrating antennas onto platforms verify installed performance of the antenna through simulations.
Performance of the antenna is much different when installed on real-world platforms than when installed on a ground plane in an anechoic chamber. Further, coupling between pairs of antenna can be radically different depending upon where the antennas are installed on the platform. Simulation thus helps to predict how an antenna will perform when installed on a car, ship, aircraft or tower.
When antennas are closely located, insignificant emissions or susceptibilities of transmitters and receivers can result in interference due to high coupling levels. Further, RF transmitter power levels, modulation schemes, harmonics and nonlinear behaviour lead to spectral growth both in-band and out-of-band that can overload receivers. Such study can be validated only through quality simulations.
The main objectives of the simulation training are:

- To familiarize industry standard design and simulation environment
- To analyse electromagnetic structures like antennas and other microwave devices and circuits

- To improve design efficiency of RF and Microwave structures
- To learn to optimize parameters for various antennas and microwave circuits
- To complement the theory and problem solving done in the coursework of Microwave Engineering and Antenna Theory and Design in higher semesters

Speaker/Trainer

Shri. Shashikumar R, Senior RF and Antenna Engineer is an industry expert who will be delivering the hands-on session, facilitated by Shri. Biju Paul, Entuple Technologies.

Participants

Targeted participants are UG and PG students, Research Scholars and Faculty members.

Important Information for Participants:

- The workshop will be conducted in online mode
- Registration is free for all participants
- The participant must download the research/student version of Ansys/HFSS personally for the hand-on session

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